ALICE L WALTON SCHOOL OF MEDICINE

# Dyslipidemia: Pathophysiology

Last updated February 12, 2026 · 16 min read · ScholarRx

Adapted by: Diptiman Bose, Phil Riley, Trager Hintze, Lance Bridges, Kimberly Keller, Marie Berenguer

Editors: Jordan Finn, Russell Johnson

Adapted from: Primary Lipid Disorders

Listen to this Brick

0:00 / 28:37 1x

Learning Objectives (3) Versions

*After completing this brick, you will be able to:*

1. Distinguish between primary and secondary lipid disorders.
2. Classify primary dyslipidemias based upon the lipoprotein disrupted, genetic inheritance pattern, and/or elevated lipids.
3. Interpret clinical and laboratory findings to properly diagnose dyslipidemia.

**CASE CONNECTION**

JA is a 36-year-old male you are seeing in the emergency department for acute left upper quadrant abdominal pain. His lipase levels are four times normal, and you diagnose acute pancreatitis. As you run through the differential for possible causes, nothing jumps out. "Check a lipid profile," the attending suggests. "I'm confused," you say. "This isn't a heart issue, it's a GI issue. Why would I check a lipid profile?"

What diagnosis is the attending pursuing? Consider your answer as you read, and we'll revisit JA at the end of the brick.

GO TO CONCLUSION ↓

## What is dyslipidemia? (LO#1)

Lipid disorders are abnormalities in lipids (fats) or lipoproteins in the blood. These dyslipidemias play an important role in cardiovascular disease. Lipid disorders are very common, as evidenced by statins (lipid-lowering drugs) being the second most commonly prescribed drug in the United States.

Many cases of dyslipidemia are idiopathic. Those with known causes are classified into primary and secondary lipid disorders. By far, the most common are the **secondary lipid disorders.** When taking a history from patients, questions that elucidate the presence of these patient factors are important to individualizing your [illegible] elevations of cholesterol and triglycerides (TGs) due to patient [illegible]:

- Diabetes mellitus
- Chronic kidney disease
- Liver disease
- Hypothyroidism
- Obesity
- Alcohol consumption
- Medications (the following are examples):
- Hydrochlorothiazide
- β-blockers
- Anti-retrovirals
- Oral contraceptives

Most of these risks are from changes in either liver or glucose metabolic pathways. All patients with elevated blood lipids should have these secondary conditions ruled out first. Diagnostic testing to exclude secondary dyslipidemias include:

- hemoglobin A1c (diabetes)
- BUN (blood urea nitrogen), sCr (creatinine) and a UA (urinalysis) for chronic kidney disease
- LFTs (liver function tests) for liver disease
- thyroid-stimulating hormone (TSH) for hypothyroidism
- BMI (body mass index) calculation for obesity

Much less common are the **Primary Lipid Disorders**, which are the focus here. These are inherited disorders that directly cause defects in lipid metabolism. Patients often have a **family history of early-onset cardiovascular disease**. We'll discuss a few of these disorders in which specific genetic causes have been identified.

Which is more common, primary or secondary dyslipidemia?

## Classify primary dyslipidemias based upon the lipoprotein disrupted, genetic inheritance pattern, and/or elevated lipids (LO#2)

One might assume that primary dyslipidemias arise from abnormally increased synthesis of these substances (e.g. cholesterol). Instead, these are more often abnormalities of lipoprotein structure or of the lipoprotein receptors, leading to **decreased lipid clearance**. Therefore, a thorough review of the identity and function of lipoproteins is required to fully understand the primary dyslipidemias (Review Brick).

### Fredrickson Classification of Primary Dyslipidemias

The historical classification of the primary disorders is the Fredrickson phenotype which was covered in BOOM (Review Brick). This system sorts disorders based on **the identity and type of lipoproteins** elevated. USMLE board exams will typically use the familial name, but for clinical knowledge it is good to be familiar with both designations. Many reference sources will use both names.

The focus for this content is on:

- **Which lipoprotein is elevated**
- **Which gene is involved (e.g., *LPL*, *LDLR*, *ApoE*)**
- **What basic inheritance patterns are observed**
- **What are the clinical consequences (pancreatitis, xanthomas, increased risk of ASCVD)**

Each numbered type represents a pattern of lipoprotein elevation:

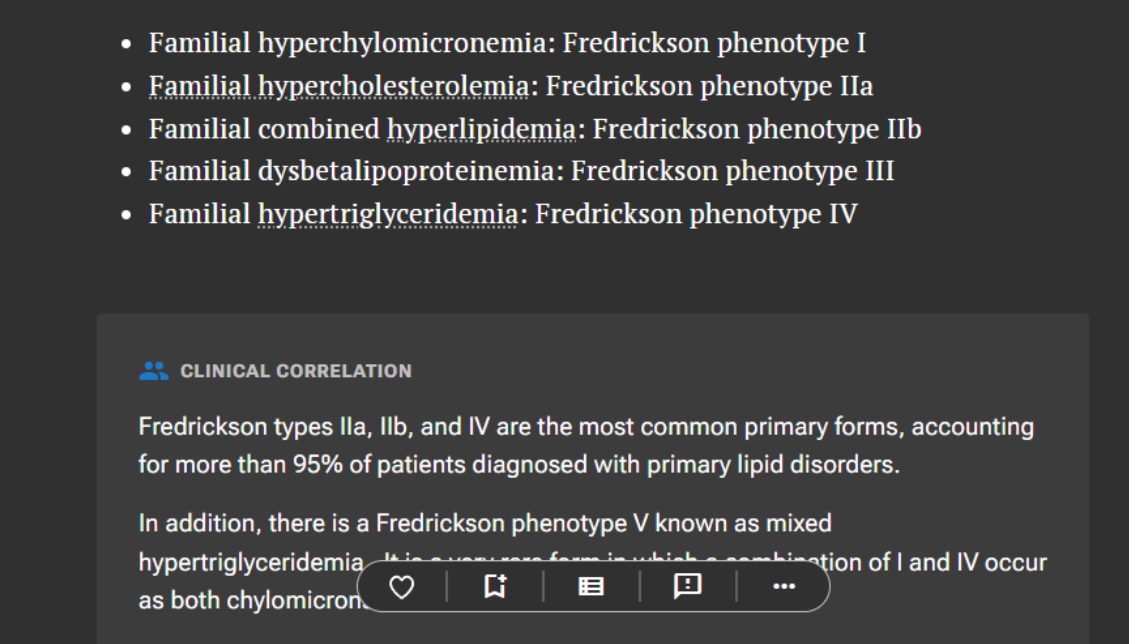
- Familial hyperchylomicronemia: Fredrickson phenotype I
- Familial hypercholesterolemia: Fredrickson phenotype IIa
- Familial combined hyperlipidemia: Fredrickson phenotype IIb
- Familial dysbetalipoproteinemia: Fredrickson phenotype III
- Familial hypertriglyceridemia: Fredrickson phenotype IV

**CLINICAL CORRELATION**

Fredrickson types IIa, IIb, and IV are the most common primary forms, accounting for more than 95% of patients diagnosed with primary lipid disorders.

In addition, there is a Fredrickson phenotype V known as mixed hypertriglyceridemia. It is a very rare form in which a combination of I and IV occur as both chylomicron[illegible]

A summary of the primary dyslipidemia with their clinical and board relevant features is provided below.



| Disease name and Fredrickson phenotype | Genetics (pattern and most common gene if known) | Lipoprotein or Apolipoprotein abnormality | Elevated lipoprotein(s) | Cholesterol levels | TG levels |
|---|---|---|---|---|---|
| Familial hyperchylomicronemia (type I) | AR<br>gene: *LPL*<br>(~80% of cases) | ↓ LPL or ↓ Apo C-2 | Chylomicrons | Normal to very mildly ↑ | ↑↑↑↑ |
| Familial hypercholesterolemia (type IIa) | AD<br>gene: *LDLR* | ↓ LDL receptors | LDL | ↑↑↑ | Normal to very mildly ↑ |
| Familial combined hyperlipidemia (type IIb) | Polygenic/Multifactorial | ↑ Apo B-100 | LDL and VLDL | ↑↑ | ↑↑ |
| Familial dysbetalipoproteinemia (type III) | AR<br>gene: *APOE* | Apo E binding defect | IDL and chylomicron remnants | ↑↑ | ↑↑ |
| Familial hypertriglyceridemia (type IV) | AD and polygenic | VLDL overproduction by liver | VLDL | Normal to very mildly ↑ | ↑↑↑ |

**Autosomal Recessive (AR); Autosomal Dominant (AD), Triglycerides (TG); Lipoprotein Lipase (LPL); Low-Density Lipoprotein Receptor (LDLR); Low-Density Lipoprotein (LDL); Intermediate-Density Lipoproteins (IDL); Very-Low-Density Lipoprotein (VLDL)**

**CLINICAL CORRELATION**

Patients may be unaware of a primary dyslipidemia. Such conditions are commonly discovered during routine screening bloodwork or when assessing for possible causes of a stroke or heart attack. The most common symptoms are due to complications the[illegible] in coronary artery disease) or abdominal pain (from pancreatitis secondary to high TG levels).

Each familial disorder has a different molecular basis although some of the unique disruptions results in similar impacts on lipid metabolism. Recognize these distinctions and commonalities (**Figure 1**).

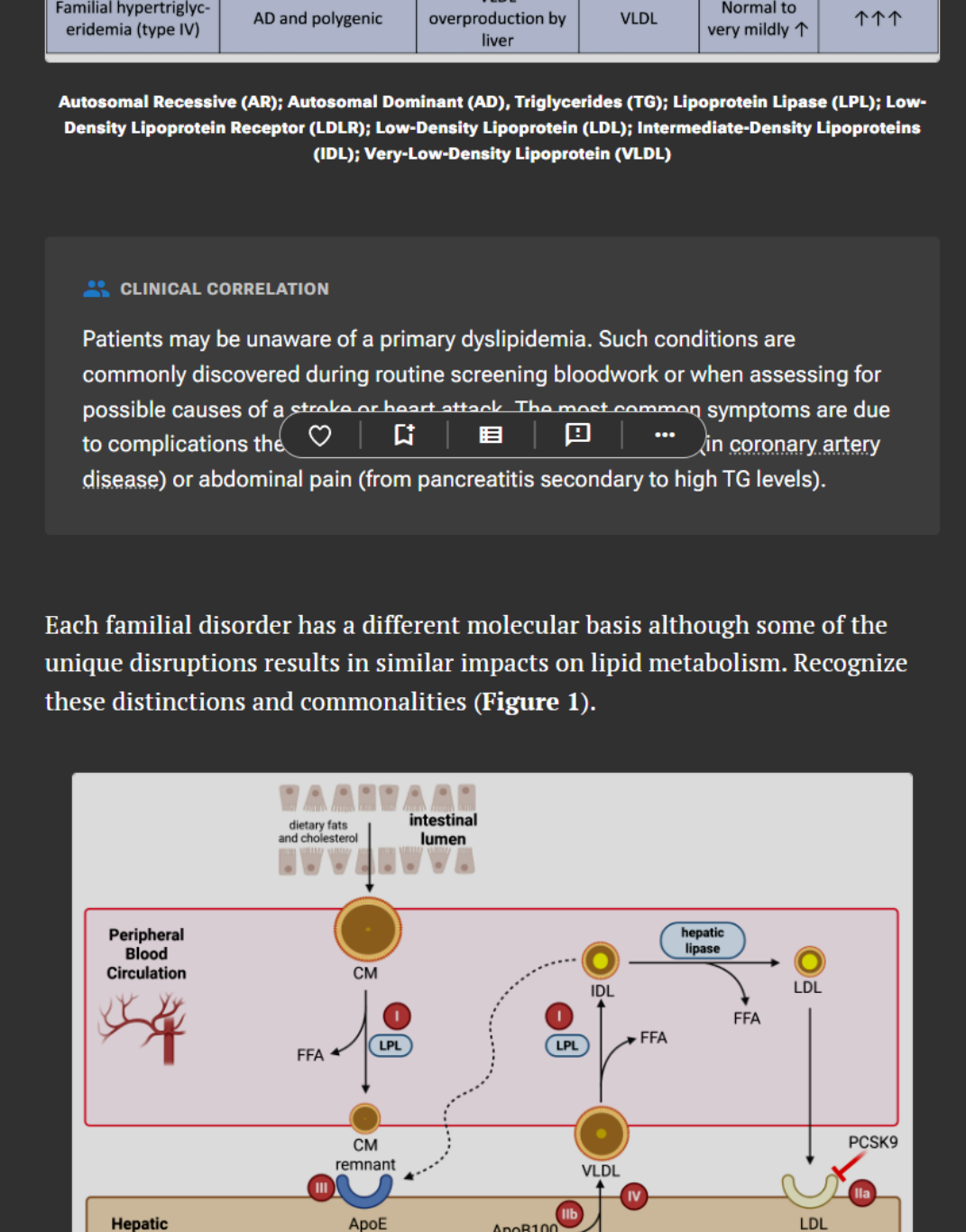


**Figure 1. Molecular basis of primary dyslipidemias.** The Fredrickson phenotype associated with a particular disruption in lipid metabolism is indicated by the red circles. I is compromised LPL enzymatic function causing retention of triglycerides within CM and VLDL particles. IIa results from decreased LDLR abundance. This could be due to issues in receptor expression itself or increased turnover of LDLR by dysregulated PCSK9. IIb and IV both lead to augmented VL[illegible] mechanisms. IIb is a result from increased ApoB100 [illegible] III results from decreased abundance of ApoE receptor that is required for the uptake of CM remnants and to a lesser extent IDL particles. chylomicron (CM), free fatty acid (FFA), lipoprotein lipase (LPL)

**Familial Hyperchylomicronemia** (Fredrickson phenotype I).

- **Deficiency in Lipoprotein Lipase (LPL).** This enzyme sits mainly on the surface of endothelial cells and degrades circulating TG's (mainly in VLDL and chylomicrons) to free fatty acids.
- **Autosomal Recessive (predominately).**
- Leads to **very high levels of TGs** (often >1,000 mg/dL), with near normal cholesterol levels. Patients are primarily at risk for **pancreatitis**, as also occurs with familial hypertriglyceridemia. Unlike many of the other forms described below, the risk for ***ischemic heart disease is not increased*** as individuals exhibit cholesterol within or near reference ranges
- Other genes can be involved such as, among others, *APOC2* encoding for Apo C-II, a protein required to activate LPL.
- Extrinsic factors (e.g. diet) influence phenotype.

**Familial Hypercholesterolemia** (Fredrickson phenotype IIa).

- Caused by specific gene mutations that lead to **too few or a complete absence of LDL receptors.** Mutations in *LDLR*, encoding the LDL receptor, are most common. These mutations alter expression **not** binding affinity.
- Only familial form that is predominately **Autosomal Dominant**. One of the most common autosomal dominant disorder in humans.
- **Very high circulating LDL levels** (even at birth). TG levels are near normal.
- Heterozygotes have increased risk of severe atherosclerotic disease in young adulthood
- Homozygous patients (uncommon) have more severe phenotype and earlier onset: often die of cardiac disease before age 20 (not a full dominance disease).
- Other genes are also involved such as *PCSK9*. Increased **PCSK9** activity leads to augmented LDLR internalization and decreased expression. **Apo B-100** is also commonly disrupted as this prevents internalization of cholesterol rich components.
- Extrinsic factors (environment) influence phenotype.

What is the mechanism for the high LDL levels in patients with familial hypercholesterolemia?

**Familial Combined Hyperlipidemia** (Fredrickson phenotype IIb, also called mixed hyperlipidemia)

- Traditionally been considered the **most common primary lipid disorder**, seen in 1% to 2% [illegible]
- Complex genetics: **Polygenic/multifactorial disorder**, resulting with variable expressivity or different subtypes of the disease.
- Non-mendelian inheritance- Males are more commonly affected than Females.
- Complex physiology with distinct abnormalities in lipid metabolism.
- Most prevalent forms entail **increased apolipoprotein B-100**, associated with hepatic overproduction of small VLDL particles as well as small, dense LDL particles
- Also seen can be hepatic overproduction of TG's, decreased hepatic clearance of LDL, chylomicron remnants and VLDL, and decreased lipoprotein lipase (LPL) activity
- Patients have moderately high levels of both cholesterol and TGs, and, basically, all the bad lipids are elevated: labs will likely show **increased levels of LDL, VLDL, and TG's** (hence the name "combined" hyperlipidemia)
- The small LDLs are especially atherogenic, and this disorder accounts for as many as 20% of cases of **premature coronary artery disease**. Patients also demonstrate insulin resistance and obesity, looking a bit like patients with metabolic syndrome or type 2 diabetes.

What lipids are elevated in familial combined hyperlipidemia?

**Familial Dysbetalipoproteinemia** (Fredrickson phenotype III)

- Primarily caused by a gene **mutation in the *APOE*** gene encoding for the **ApoE lipoprotein**
- **Autosomal recessive** mainly. In rare cases, heterozygous mutations in the *APOE* gene have been reported, following an autosomal dominant pattern and leading to earlier onset with a more severe phenotype.
- ApoE is on the surface of LDL, chylomicron remnants, and VLDL and is needed for entry of these lipid particles to the liver for metabolism (and eventual clearance). The lack of entry means that chylomicron remnants, VLDLs, and IDLs remain in circulation
- Moderate elevations in total cholesterol and TGs
- Increased risk for premature heart disease
- Palmar xanthomas are pathognomonic (see below for xanthoma description).
- Extrinsic factors (environment) influence phenotype.

**CLINICAL CORRELATION**

The ApoE isoform ApoE2 is clinically significant due to its reduced affinity for LDL receptor. Poor binding of this isoform decreases hepatic clearance of chylomicrons and I[illegible] levels. Additionally, the ApoE4 isoform is strongly associated with Alzheimer's disease.

**Familial Hypertriglyceridemia** (Fredrickson phenotype IV)

- **Pure Familial Hypertriglyceridemia:** a **very rare** monogenic form with **Autosomal dominant inheritance,** caused mostly by mutations in ***LPL* gene**, critical for clearing triglyceride-rich lipoproteins
- **Most cases are polygenic/ multifactorial,** with the phenotype influenced by the interaction of genetic variants from over 30 genes and environmental factors such as obesity, insulin resistance, diabetes, alcohol intake, and oral estrogen use. This leads to variable expressivity with diverse metabolism signature and clinical presentation.
- Due to hepatic **overproduction of VLDLs** the lipid panel will show **elevated VLDL and TG**. Total cholesterol is relatively normal, and the HDL is often low. Because the LDL levels are not as increased, the association between this disease and atherosclerotic heart disease is not as strong as other primary dyslipidemias but it is a complication.
- Often associated with type 2 diabetes mellitus
- Increased risk for **acute pancreatitis** due to high TG levels

High levels of [illegible] pancreatitis?

**CLINICAL CORRELATION**

The supernatant of spun blood in hypertriglyceridemia and hyperchylomicronemia is creamy or milky due to the high triglyceride content.

To keep track of the categories of primary lipid disorders use the mnemonic below.

Mnemonic

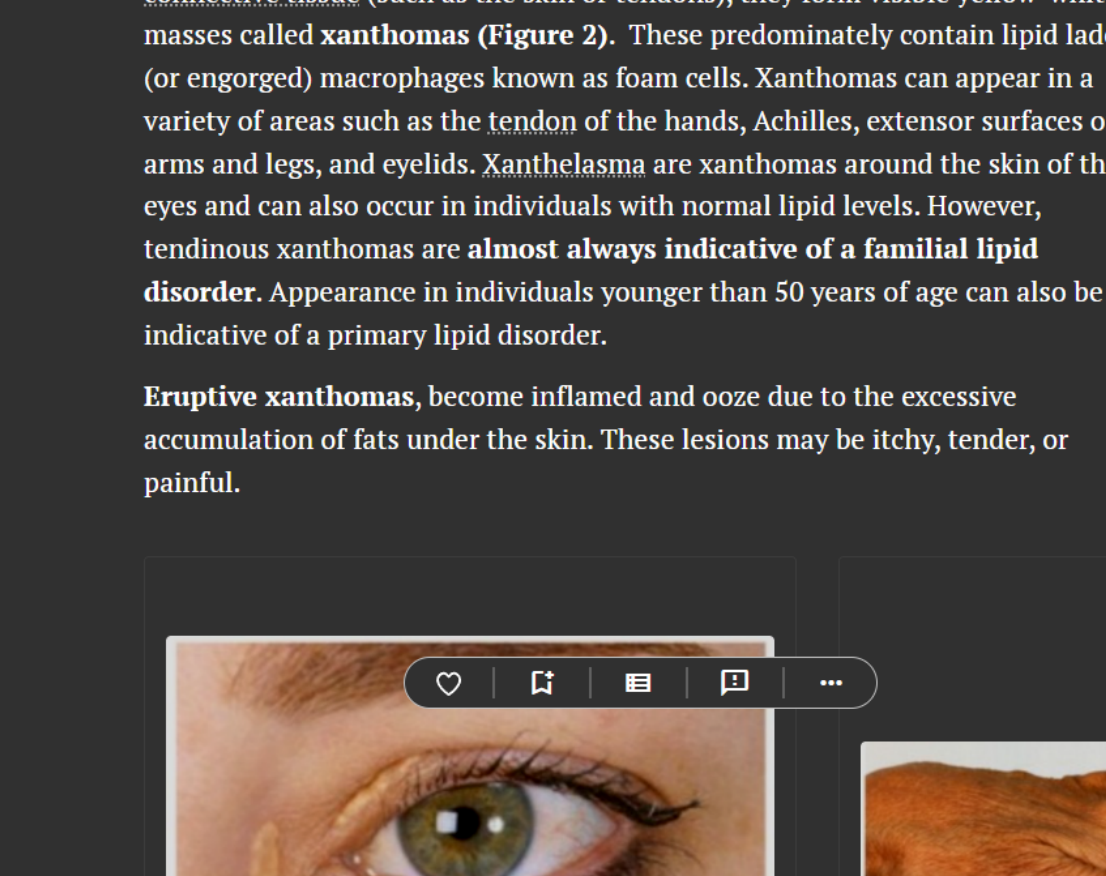
**1 LP, 2 LD, 3 with E, 4 gets more:**
Type **1** (familial hyperchylomicronemia) is a **lipoprotein lipase (LP**L) deficiency.
Type **2** (familial hypercholesterolemia) is an **LD**L receptor deficiency.
Type **3** (familial dysbetalipoproteinemia) is due to an ***APOE*** variant.
Type **4** (familial hypertriglyceridemia) gets **more** VLDL production (also VLDL has 4 letters).

### Clinical Indicators of Primary Lipid Disorders: Family History, Xanthomas, Corneal Arcus (LO#3)

Patients with primary disorders will often have a history of family members with early onset cardiovascular disease or pancreatitis. If primary disorders are suspected, the physical exam and labs will provide clues. In addition, some of these disorders also lead to characteristic physical exam findings.

When lipid products accumulate in macrophages in the subepithelial connective tissue (such as the skin or tendons), they form visible yellow-white masses called **xanthomas (Figure 2)**. These predominately contain lipid laden (or engorged) macrophages known as foam cells. Xanthomas can appear in a variety of areas such as the tendon of the hands, Achilles, extensor surfaces of arms and legs, and eyelids. Xanthelasma are xanthomas around the skin of the eyes and can also occur in individuals with normal lipid levels. However, tendinous xanthomas are **almost always indicative of a familial lipid disorder**. Appearance in individuals younger than 50 years of age can also be indicative of a primary lipid disorder.

**Eruptive xanthomas**, become inflamed and ooze due to the excessive accumulation of fats under the skin. These lesions may be itchy, tender, or painful.



*2012 Bell et al., Dove Medical Press Ltd.*

Xanthelasma: eyelid plaques. Please NOTE these are not unique to dyslipidemias and can occur due to conditions other than abnormal lipid accumulation

*2012 Bell et al.,*

Exten

1 / 3

**Figure 2. Xanthomas in distinct bodily locations.**

In addition to dermatological signs, excessive lipid deposition within the eye can appear as a white corneal arcus (**Figure 3**). These are predominately associated with cholesterol levels but can also involve excess triglycerides. These can also appear in older individuals (> 50 years of age) and may, at that stage, be a benign sign of aging.

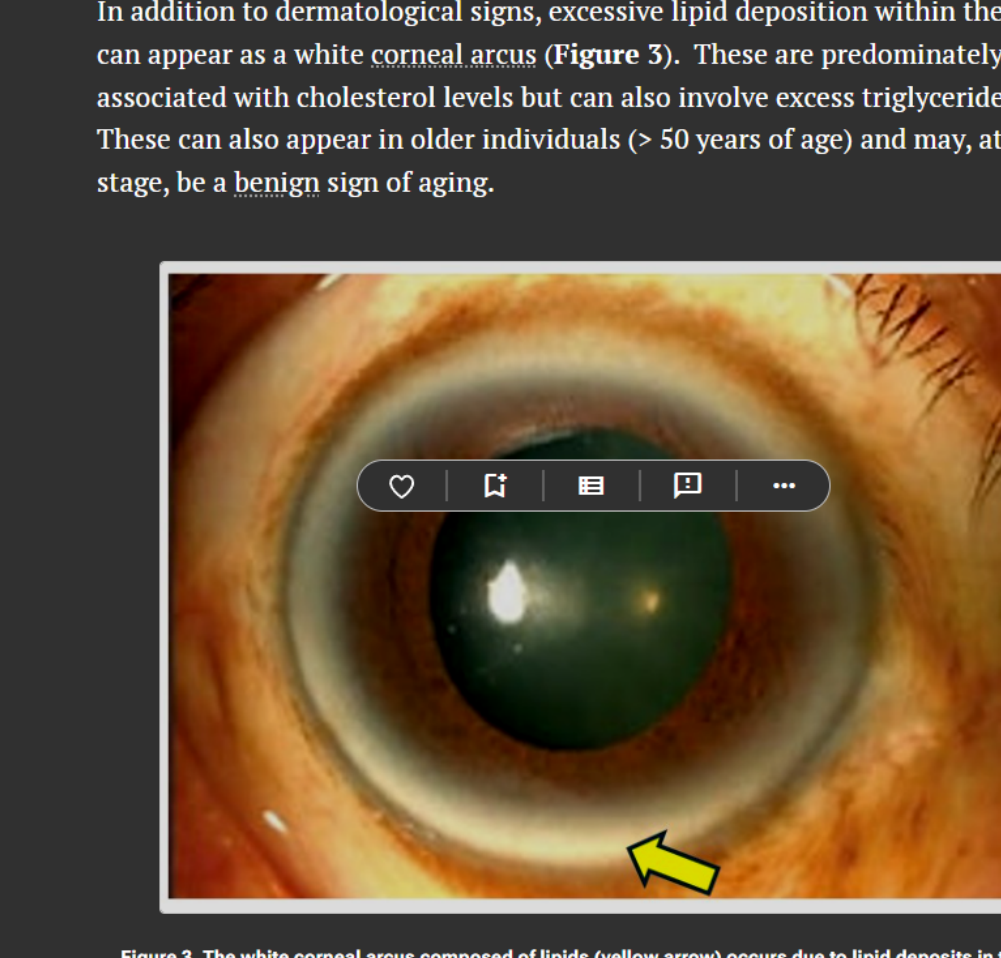


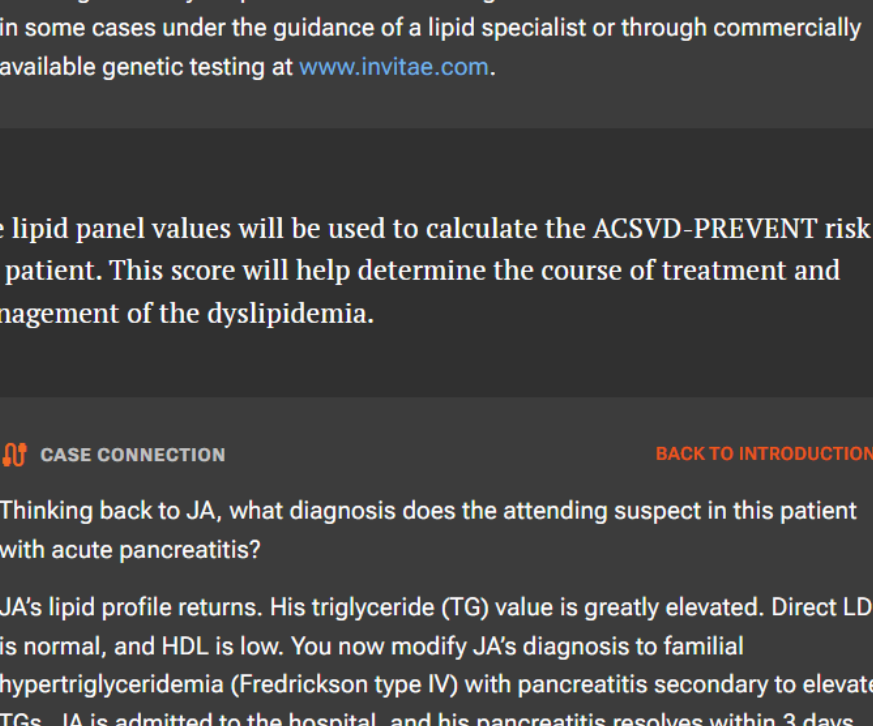
**Figure 3. The white corneal arcus composed of lipids (yellow arrow) occurs due to lipid deposition in the iris. Corneal arcus before ~40 years of age is highly suggestive of familial hypercholesterolemia.**

**Lipemia retinalis** occurs in patients with excessively high TG levels (> 2,000 mg/dL). On fundoscopic exam, the retinal arteries appear yellow or white rather than red because of an excessive accumulation of lipids.

**CLINICAL CORRELATION**

With respect to the xanthomas, **palmar xanthomas**, distinctive yellow-orange plaques that appear along the palmar creases of the hands, are uniquely **pathognomonic** for **type III: familial dysbetalipoproteinemia**.

### Laboratory Tests (LO#3)



Flabs

Hello@flabs.in
+91 7253928905
https://www.flabs.in/

Name : Mr Dummy
Age/Gender : 20/Male
Referred By : Self
Phone No. :

Patient ID : PN2
Report ID : RE1
Collection Date : 24/06/2023 08:49 PM
Report Date : 24/06/2023 09:02 PM

BIOCHEMISTRY
LIPID PROFILE

| TEST DESCRIPTION | RESULT | REF. RANGE | UNIT |
|---|---|---|---|
| Total Cholesterol | 156 | 0 - 200 | mg/dl |
| Triglycerides level | 150 | 0 - 170 | mg/dl |
| HDL Cholesterol | 45 | 40 - 70 | mg/dl |
| LDL Cholesterol | 81.00 | 0 - 100 | mg/dl |
| VLDL Cholesterol | 30.00 | 6 - 38 | mg/dl |
| **LDL/HDL RATIO** | **1.80** | **2.5 - 3.5** | |
| **Total Cholesterol/HDL RATIO** | **3.47** | **3.5 - 5** | |

Interpretation:

| Lipid Profile Test | Desirable Levels | Borderline High | High |
|---|---|---|---|
| Total cholesterol | <200 mg/dL | 200-239 mg/dL | ≥240 mg/dL |
| LDL cholesterol | <100 mg/dL | 130-159 mg/dL | ≥160 mg/dL |
| HDL cholesterol | ≥60 mg/dL | 40-59 mg/dL | <40 mg/dL |
| Triglycerides | <150 mg/dL | 150-199 mg/dL | ≥200 mg/dL |

Desirable levels of cholesterol and triglycerides are associated with a lower risk of heart disease, while high levels increase the risk. HDL cholesterol is often called "good" cholesterol, as it helps to remove excess cholesterol from the blood vessels. In contrast, LDL cholesterol is often called "bad" cholesterol, as it contributes to the buildup of plaque in the arteries.

**Sample Lipid Profile showing measured and reference range values**

A fasting lipid panel is the most crucial test to determine which specific levels are elevated and the exact disorder subtype. This consists of total cholesterol, VLDL, LDL, and HDL cholesterol levels and TG's (see sample labs above).

The **LDL is calculated** using the Friedewald equation (see below):

Calculated LDL = Total Cholesterol - HDL - (Triglycerides ÷ 5)

**CLINICAL CORRELATION**

Non-fasting TG's (which are usually high) or genetically elevated TG levels cause incorrect LDL calculations. You can request a "Direct LDL" measurement when ordering labs for your patient. Genetic testing for characteristic mutations is done in some cases under the guidance of a lipid specialist or through commercially available genetic testing at www.invitae.com.

The lipid panel values will be used to calculate the ACSVD-PREVENT risk for the patient. This score will help determine the course of treatment and management of the dyslipidemia.

**CASE CONNECTION**

BACK TO INTRODUCTION ↑

Thinking back to JA, what diagnosis does the attending suspect in this patient with acute pancreatitis?

JA's lipid profile returns. His triglyceride (TG) value is greatly elevated. Direct LDL is normal, and HDL is low. You now modify JA's diagnosis to familial hypertriglyceridemia (Fredrickson type IV) with pancreatitis secondary to elevated TGs. JA is admitted to the hospital, and his pancreatitis resolves within 3 days. You explain to JA that he should take a medication to lower his TG level. In a 6-month follow-up visit, he reports no further episodes of abdominal pain. "Whatever that medication did to my triglycerides, I'm happy with the outcome. I haven't had any abdominal pain," JA says.

## Summary

- Lipid disorders are often secondary to conditions such as diabetes, kidney disease, or alcohol use, but may also result from inherited abnormalities in lipid metabolism, with primary dyslipidemias classified by Fredrickson phenotypes
- Dyslipidemias that lead to high low-density lipoprotein (LDL) cholesterol cause early or accelerated atherosclerosis, including early-onset coronary artery disease (chest pain).
- Patients with primary lipid disorders that cause elevated triglycerides (TGs) are at increased risk for pancreatitis.
- Familial combined hyperlipidemia (type IIb) is most common and leads to elevated LDL, very low-density lipoproteins (VLDLs), and TGs due to multiple defects in lipid clearance and synthesis.
- Familial hypercholesterolemia (type IIa) is caused by mutations to the LDL receptor gene.
- Familial hypertriglyceridemia (type IV) and familial hyperchylomicronemia (type I) have elevated TGs and do not have increased risk of atherosclerotic heart disease; however, they both have increased risk for recurrent pancreatitis.
- Familial hypertriglyceridemia (type IV) is characterized by elevated TGs due to VLDL overproduction.
- Physical exam findings such as xanthomas and corneal arcus may offer clues that a patient has a primary lipid disorder.
- A lipid panel determines which specific levels are elevated and the exact disorder subtype.

## Review Questions

**1. Which of the following primary lipid disorders would most likely lead to acute pancreatitis?**

A. ○ Defective low-density lipoprotein (LDL) receptors
B. ○ Familial dysbetalipoproteinemia
C. ○ Familial combined hyperlipidemia
D. ○ Familial hypercholesterolemia
E. ● Familial hypertriglyceridemia

**Hide Explanation**

The correct answer is familial hypertriglyceridemia (E). The primary lipid disorders that most likely cause pancreatitis are those that lead to marked elevations in triglyceride (TG) levels (ie, familial hypertriglyceridemia and familial hyperchylomicronemia—rare and caused by lipoprotein lipase deficiency). Defective low-density lipoprotein (LDL) receptors (A) are more likely to show high LDL cholesterol and predispose to accelerated atherosclerosis. Neither familial dysbetalipoproteinemia (B) nor familial combined hyperlipidemia (C) cause severe pancreatitis. Familial hypercholesterolemia (D) is caused by defective LDL receptors.

**2. An 8-year-old male presents to his primary care provider with recurrent bouts of abdominal pain, especially after eating. The patient reports no recent sick contac[illegible] seafood. His family history reveals that a cousin had a genetic condition that required him to follow a strict diet. On exam, the physician notes many reddish bumps over the patient's buttocks and back. The patient's liver is palpable. He has reduced sensation over his feet and ankles bilaterally. The lipid panel shows the following results: total cholesterol, 200 mg/dL; LDL cholesterol, 87 mg/dL; HDL cholesterol, 50 mg/dL; and triglycerides 3156 mg/dL. Which defective protein is the most likely cause of this patient's condition?**

A. ○ Apolipoprotein B
B. ○ Apolipoprotein E2
C. ○ HMG-CoA reductase
D. ○ LDL receptor
E. ● Lipoprotein lipase

**Hide Explanation**

The correct answer is lipoprotein lipase (E). This patient presents with findings consistent with familial hyperchylomicronemia (eg, elevated TG levels >2000 mg/dL, relatively normal cholesterol levels, abdominal pain indicative of pancreatitis, hepatomegaly, eruptive xanthomas, and peripheral neuropathy). This rare disorder is most commonly caused by a deficiency in lipoprotein lipase (familial hyperchylomicronemia). An apolipoprotein B (A) defect causes familial hypercholesterolemia, which presents with elevated low-density lipoprotein (LDL) cholesterol. Patients [illegible] subtype have familial dysbetalipoproteinemia, which does not tend to present with hepatomegaly or pancreatitis. HMG-CoA reductase (C) is the key enzyme in cholesterol biosynthesis and the target of statin drugs, but mutations of this enzyme are not a cause of these disorders. Like an ApoE defect, an LDL receptor (D) deficit causes familial hypercholesterolemia.

**3. A 38-year-old female presents to her primary care provider noting an increasing number of large, bumpy growths over her knees and buttocks as well as yellow streaks in the creases of her palms. Both her mother and father developed heart disease before age 40 years. The serum cholesterol and triglycerides are both elevated. Which of the following primary lipid disorders is most likely?**

A. ○ Familial combined hyperlipidemia
B. ● Familial dysbetalipoproteinemia
C. ○ Familial hypercholesterolemia
D. ○ Familial hyperchylomicronemia
E. ○ Familial hypertriglyceridemia

**Hide Explanation**

The correct answer is familial dysbetalipoproteinemia (B). The patient is describing eruptive xanthomas [illegible] pathognomonic for this autosomal recessive disease (note the early heart disease in both mother and father, suggesting accelerated atherosclerosis). The disorder is caused by ineffective clearance of chylomicron and very low-density lipoprotein remnants due to an *APOE* genetic variant, leading to elevated cholesterol and triglycerides (TGs). The other listed diseases do not cause eruptive or palmar xanthomas. Familial combined hyperlipidemia (A) causes moderate elevations in cholesterol and triglycerides. Familial hypercholesterolemia (C) causes elevated cholesterol but normal TGs. Familial hyperchylomicronemia (D) and familial hypertriglyceridemia (E) both present with normal cholesterol but elevated TGs.

**4. A 20-year-old male presents to the student health clinic for a routine wellness examination. He reports no current symptoms but admits to a diet high in fried foods and limited physical activity. His weight is stable, BMI is 26 kg/m², and blood pressure is 128/82 mmHg. A fasting lipid panel is obtained: Total Cholesterol: 210 mg/dL Triglycerides: 180 mg/dL HDL-Cholesterol: 40 mg/dL. Using the Friedewald equation, what is this patient's calculated LDL-cholesterol?**

A. ○ 110 mg/dL
B. ○ 124 mg/dL
C. ● 134 mg/dL
D. ○ 150 mg/dL
E. ○ 178 mg/dL

**Hide Explanation**

The correct answer is 134 mg/dL (C). Using the formula, LDL = Total Cholesterol - HDL - (Triglycerides ÷ 5) calculate the LDL-C level. The remaining answers (A, B , D ,E) are incorrect calculations.

**How would you rate this Brick?**

## References

- Hill MF, Bordoni B. Hyperlipidemia. In: *StatPearls*. StatPearls Publishing. Updated February 8, 2022. Accessed September 29, 2022. https://www.ncbi.nlm.nih.gov/books/NBK559182/#_article-23187_s3

**Review Learning Objectives**

Check off the objectives when you feel comfortable with the concept

01 ☑ Distinguish between primary and secondary lipid disorders.

## References

- Hill MF, Bordoni B. Hyperlipidemia. In: *StatPearls*. StatPearls Publishing. Updated February 8, 2022. Accessed September 29, 2022. https://www.ncbi.nlm.nih.gov/books/NBK559182/#_article-23187_s3

**Review Learning Objectives**

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Distinguish between primary and secondary lipid disorders.~~

02 ☑ ~~Classify primary dyslipidemias based upon the lipoprotein disrupted, genetic inheritance pattern, and/or elevated lipids.~~

03 ☑ ~~Interpret clinical and laboratory findings to properly diagnose dyslipidemia.~~

Objective Confidence: 3/3